

Vrajesh Patel

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HIGHLIGHT OF QUALIFICATIONS

- **Results-driven Data Engineer** with 3+ years of experience designing and scaling **financial and regulatory data platforms** that power revenue, compliance, and analytics initiatives.
- Expertise in building **ETL frameworks and workflow automation** with Airflow, Databricks, and Delta Lake — improving data quality, lineage visibility, and audit compliance.
- Recognized by RBC with multiple **Performance Awards** for innovation, technical excellence, and impact in data engineering transformation initiatives.

TECHNICAL SKILLS

Programming Languages: Python, MSSQL, NoSQL, Shell

BigData Tools: Hadoop, Kafka Streaming, Spark, Delta Lake

Software & Tools: Azure, AWS, Azure Databricks, Apache Airflow, Git, Docker, OCP, ELK, Power BI, Tableau, Jupyter

Analytics Libraries: PySpark, Pandas, NumPy, Matplotlib, Seaborn, Plotly, SAP BI Power Tool

WORK EXPERIENCE

Royal Bank of Canada

Senior Data Engineer | May 2025 – Present

- Designed and implemented a **YAML-driven configuration framework** that standardized ingestion and transformation patterns across pipelines, reducing development effort and improving consistency across distributed Spark workloads.
- **Own the architecture, reliability, and operational health** of approximately **100 production data** pipelines supporting risk analytics and executive reporting, operating under strict correctness, availability, and audit requirements.
- Re-architected **CI CD processes by migrating from Jenkins to GitHub Actions**, introducing reusable workflows, automated validation gates, and controlled rollback strategies, improving deployment **stability by 30 percent**.
- Engineered scalable aggregation and transformation layers using Spark and SQL, optimizing partition strategies and execution plans to **reduce downstream query latency** for both scheduled and ad hoc workloads.
- Implemented **end-to-end observability** across pipelines using SLA monitoring, lineage tracking, and proactive alerting, improving detection and resolution times during peak reporting cycles by 20 percent.
- Delivered **regulator-facing datasets** with strong guarantees around **reproducibility, traceability, and audit transparency**, contributing to clean audit cycles and zero data integrity escalations during reporting reviews.

Senior Data Engineer | Jan 2023 – May 2025

- Led migration of on-prem transactional pipelines to Azure Databricks and Apache Airflow, designing a scalable, secure cloud data platform for **high-volume, confidential financial datasets**.
- Built a configuration-driven onboarding framework using reusable YAML templates and parameterized workflows, reducing **dataset integration time by 70 percent** and enabling consistent Spark job orchestration.
- Optimized Databricks workloads by analyzing job execution patterns and implementing serverless and role-based cluster configurations, **reducing compute costs by 27 percent** while maintaining performance SLAs.
- Re-designed data flows into **Medallion Architecture** to isolate failures, improve debuggability, and prevent incomplete data from propagating downstream, resulting in a **35 percent improvement in SLA adherence**.
- Designed and maintained optimized Delta Lake tables using partitioning, compaction, and vacuum strategies to sustain performance at scale.
- Developed a **data quality and compliance framework** using Great Expectations and Soda, enforcing schema contracts, completeness thresholds, and timeliness checks across distributed data flows.
- Designed a star schema model and materialized views for BI workloads, reducing **dashboard load times by 45 percent** and improving analytical responsiveness for business stakeholders.

CERTIFICATIONS AND AWARDS

- **Azure Databricks Associate Data Engineer** - Microsoft Azure
- **RBC Annual Performance Award** - Airflow Platform Team
- **RBC Quarterly Performance Award** - Airflow Platform Team

EDUCATION

York University, Bachelor of Science, Computer Science